



Ingeniería en Ciencias de la Computación

Sistemas Distribuidos

Capturas del Empaquetamiento y Despliegue de NoticeBoard con Docker

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Índice

1. Análisis y Desarrollo Local	2
2. Construcción de Imagen y Verificación	7
3. Publicación en Docker Hub	10

1. Análisis y Desarrollo Local

```

Administrador de Símbolos del sistema python run.py
Requirement already satisfied: MarkupSafe>=2.0 in C:\Users\VSGI\KATANA\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from Jinja2>=3.1.2-->Flask>=3.0.3-->requirements.txt (line 1)) (3.0.3)
Collecting gunicorn [from sqlalchemy>=2.0.0-->Flask-SQLAlchemy>3.1.1-->requirements.txt (line 2)]
  Downloading gunicorn-21.0.0-cp314-cp314-win_msvc14_x64.whl.metadata (3.8 kB)
Collecting greenlet<3.1.0, >=3.0.0 [from sqlalchemy>=2.0.0-->Flask-SQLAlchemy>3.1.1-->requirements.txt (line 2)]
  Downloading greenlet-3.0.0-cp314-cp314-win_msvc14_x64.whl (2.6 MB)
  Downloading Flask-SQLAlchemy-3.0.3-py3-none-any.whl (181 kB)
  Downloading Flask-SQLAlchemy-3.1.1-py3-none-any.whl (25 kB)
  Downloading blinker-1.9.0-py3-none-any.whl (8.3 kB)
  Downloading itsdangerous-2.2.0-py3-none-any.whl (14 kB)
  Downloading sqlalchemy-2.0.40-cp314-cp314-win_msvc14_x64.whl (2.1 MB)
  Downloading greenlet-5.5.0-cp314-cp314-win_msvc14_x64.whl (296.0 kB)
  Downloading werkzeug-3.1.0-py3-none-any.whl (226 kB)
Installing collected packages: werkzeug, itsdangerous, greenlet, blinker, sqlalchemy, Flask, Flask-SQLAlchemy
  Running setup for Flask-SQLAlchemy 3.1.1
  WARNING: The script flask.exe is installed in 'C:\Users\VSGI\KATANA\AppData\Local\Python\pythoncore-3.14-64\Scripts' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-manifest-location
Successfully installed flask-3.0.3 flask-sqlalchemy-3.1.1 gunicorn-21.0.0 blinker-1.9.0 greenlet-5.5.0 itsdangerous-2.2.0 sqlalchemy-2.0.49

[notice] A new release of pip is available: 26.0.2 -> 26.1.1
[notice] To update, run: C:\Users\VSGI\KATANA\AppData\Local\Python\pythoncore-3.14-64\python.exe -m pip install --upgrade pip

C:\Users\VSGI\KATANA\Downloads\noticeboard\noticeboard\python run.py
 * Serving Flask app 'app'
 * Debug mode: off

WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://172.16.222.236:5000
Press Ctrl+C to quit
172.16.222.236 - [18/May/2026:10:39:57] "GET / HTTP/1.1" 200 -
172.16.222.236 - [18/May/2026:10:39:57] "GET /static/style.css HTTP/1.1" 200 -
172.16.222.236 - [18/May/2026:10:39:57] "GET /favicon.ico HTTP/1.1" 404 -
172.16.222.236 - [18/May/2026:10:40:00] "GET /category/general HTTP/1.1" 200 -
172.16.222.236 - [18/May/2026:10:40:00] "GET /static/style.css HTTP/1.1" 200 -
172.16.222.236 - [18/May/2026:10:40:00] "GET /category/products HTTP/1.1" 200 -
172.16.222.236 - [18/May/2026:10:40:00] "GET /static/style.css HTTP/1.1" 200 -
172.16.222.236 - [18/May/2026:10:40:00] "GET /category/categories HTTP/1.1" 200 -
172.16.222.236 - [18/May/2026:10:40:00] "GET /static/style.css HTTP/1.1" 200 -
172.16.222.236 - [18/May/2026:10:40:00] "GET /category/administrativo HTTP/1.1" 200 -
172.16.222.236 - [18/May/2026:10:40:00] "GET /static/style.css HTTP/1.1" 200 -
172.16.222.236 - [18/May/2026:10:40:00] "GET / HTTP/1.1" 200 -
172.16.222.236 - [18/May/2026:10:40:00] "GET /static/style.css HTTP/1.1" 200 -

```

Figura 1: Ejecución exitosa del script de inicialización en el entorno de desarrollo local.

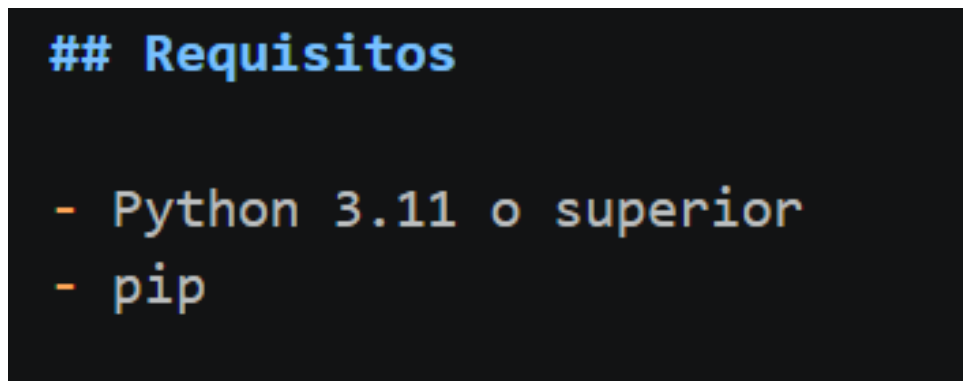


Figura 2: Identificación de la versión base de Python solicitada.

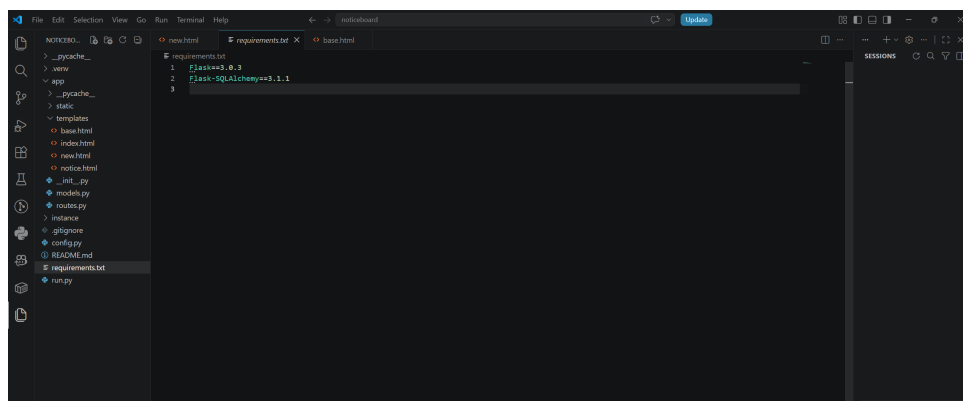


Figura 3: Revisión de dependencias (Flask y Flask-SQLAlchemy) en el archivo requirements.

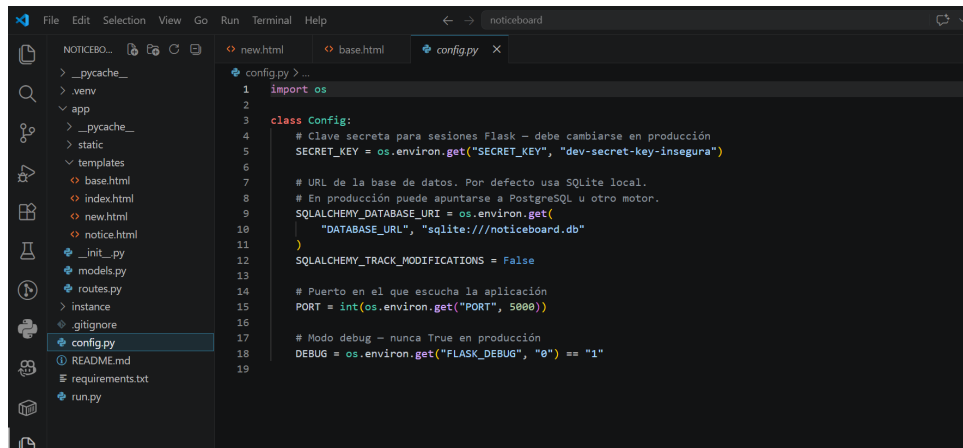


Figura 4: Análisis del archivo config.py mostrando el puerto dinámico 5000.

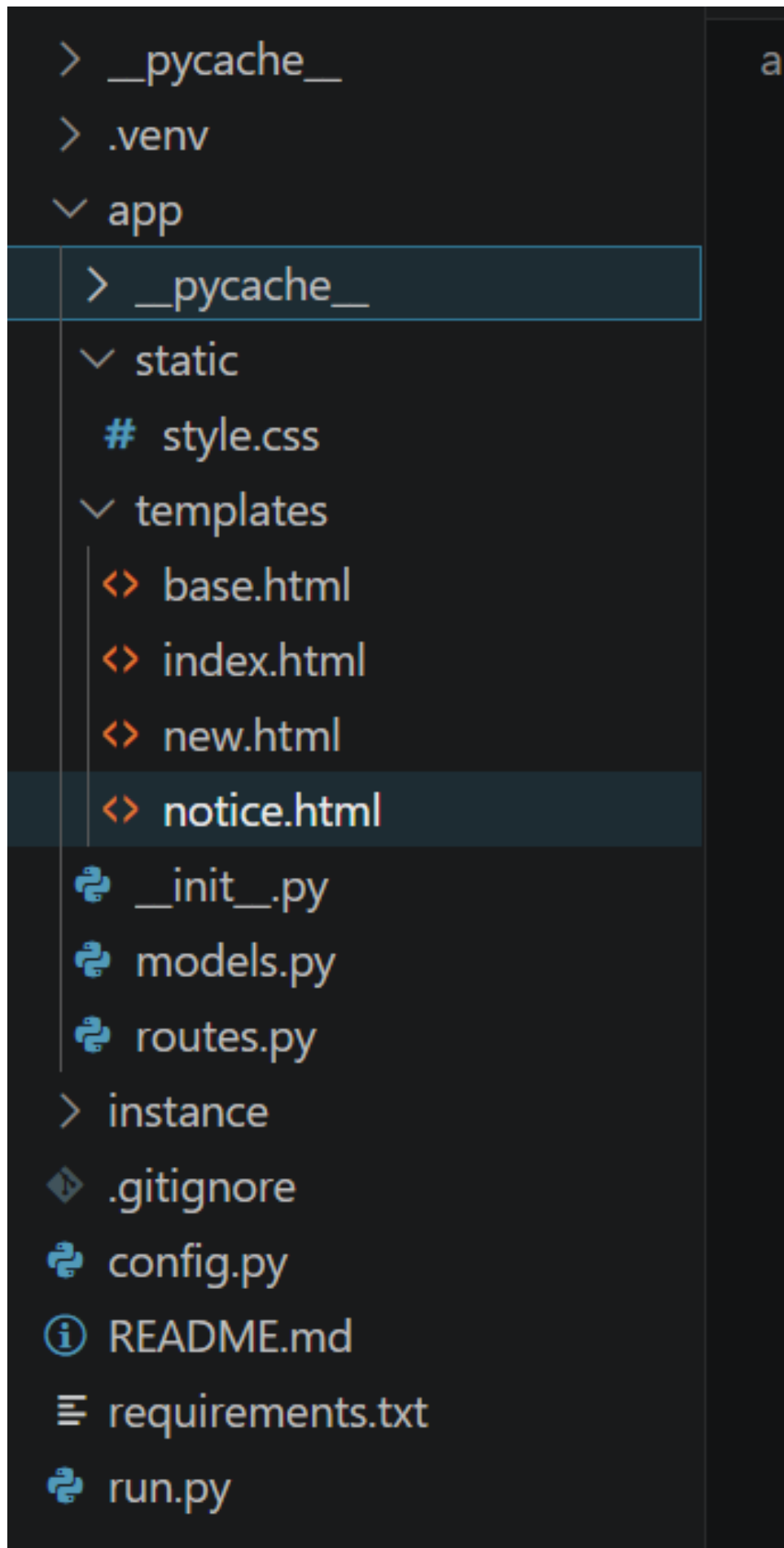


Figura 5: Estructura de directorios mostrando la ubicación de los archivos estáticos.

```

<!DOCTYPE html>
<html lang="es">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>{% block title %}NoticeBoard{% endblock %}</title>
  <link rel="stylesheet" href="{{ url_for('static', filename='style.css') }}" />
</head>
<body>

  <nav>
    <a class="brand" href="{{ url_for('main.index') }}">
      Notice<span>Board</span>
    </a>
    <a class="nav-link" href="{{ url_for('main.new_notice_form') }}">+ Nuevo anuncio</a>
  </nav>

  <div class="container">
    {% block content %}{% endblock %}
  </div>

  <footer>
    NoticeBoard · Sistemas Distribuidos · Modificado por: Jose Salamea · Universidad Politécnica Salesiana
  </footer>

</body>
</html>

```

Figura 6: Primera fase de modificaciones en el código fuente de las plantillas.

```

18   <form method="POST" action="{{ url_for('main.new_notice_submit') }}">
33   <div class="form-group">
35     <input
43     />
44   </div>
45
46   <div class="form-group">
47     <label for="category">Categoria *</label>
48     <select id="category" name="category">
49       {% for cat in categories %}
50         <option
51           value="{{ cat }}"
52           {{ 'selected' if form_data and form_data.category == cat else '' }}
53         >
54           {{ cat }}
55         </option>
56       {% endfor %}
57     </select>
58   </div>
59
60   <div class="form-group">
61     <label for="content">Contenido *</label>
62     <textarea
63       id="content"
64       name="content"
65       required
66       placeholder="Escribe aquí el cuerpo del anuncio..."
67     >{{ form_data.content if form_data else '' }}</textarea>
68   </div>
69
70   <label>Modificado por: Jose Salamea</label>
71
72   <div class="actions">
73     <button type="submit" class="btn btn-primary">Publicar anuncio</button>
74     <a href="{{ url_for('main.index') }}" class="btn btn-secondary">Cancelar</a>
75   </div>
76
77 </form>
78 </div>
79

```

Figura 7: Primera fase de modificaciones en el código fuente de las plantillas.

```

app > static > # style.css > subnav span
263 .form-group input:focus,
264 .form-group select:focus,
265 .form-group textarea:focus {
266   outline: none;
267   border-color: #2563eb;
268   background: #fff;
269 }
270
271 .form-group textarea { resize: vertical; min-height: 130px; }
272
273 /* --- Footer --- */
274 footer {
275   text-align: center;
276   padding: 2rem 1rem;
277   color: #94a3b8;
278   font-size: .82rem;
279 }
280
281 /* --- Header secundario --- */
282
283 .subnav {
284   background-color: #1a63ac;
285   padding: 8px 20px;
286   text-align: right;
287   border-bottom: 1px solid #e0e0e0;
288   box-shadow: 0 2px 4px rgba(0,0,0,.05);
289 }
290
291 .subnav span {
292   font-family: Arial, sans-serif;
293   font-size: 20px;
294   color: #d1d8d4;
295   font-weight: bold;
296 }

```

Figura 8: Modificaciones de diseño en la hoja de estilos CSS.

The screenshot shows a web application titled "NoticeBoard" with a dark blue header. In the top right corner of the header, there is a link "+ Nuevo anuncio" and a text label "Modificado por: Jose Salamea". The main content area is light gray and features a white form titled "Nuevo anuncio". The form contains the following fields: "Titulo *" with a placeholder "Escribe un título claro y breve", "Autor *" with a placeholder "Tu nombre o el del área", "Categoria *" with a dropdown menu currently showing "General", and "Contenido *" with a placeholder "Escribe aquí el cuerpo del anuncio...". Below the form, it says "Modificado por: Jose Salamea" and has two buttons: "Publicar anuncio" (blue) and "Cancelar" (gray). At the bottom of the page, there is a small footer: "NoticeBoard - Sistemas Distribuidos - Jose Salamea - Universidad Politécnica Salesiana".

Figura 9: Aplicación NoticeBoard local reflejando los cambios visuales integrados.

2. Construcción de Imagen y Verificación

```
1  .venv/
2
3  __pycache__/
4  *.pyc
5  *.pyo
6  *.pyd
7
8  instance/
9  *.db
10 *.sqlite3
11
12 .DS_Store
13 __MACOSX/
14
15 .git/
16 .gitignore
```

Figura 10: Definición de `.dockerignore` para optimizar la imagen: excluye `.venv` y cachés por eficiencia de peso, `instance/` y archivos `.db` por seguridad de datos, y carpetas `.git` para evitar arrastrar metadatos de desarrollo.


```

1 FROM python:3.11
2
3 RUN useradd -m usuario
4
5 WORKDIR /app
6
7 COPY requirements.txt .
8
9 RUN pip install -r requirements.txt
10
11 COPY . .
12
13 RUN chown -R usuario:usuario /app
14
15 USER usuario
16
17 ENV PORT=5000 \
18     FLASK_DEBUG=0 \
19     PYTHONUNBUFFERED=1
20
21 EXPOSE 5000
22
23 CMD ["python", "run.py"]

```

Figura 11: Configuración de Dockerfile: utiliza python:3.11 completo para asegurar compatibilidad de librerías, separa requirements.txt del código para optimizar la caché de capas en futuras compilaciones, y restringe permisos mediante USER usuario para mitigar riesgos de seguridad.

```

C:\Users\MSI_KATANA\Downloads\noticeboard\noticeboard>docker build -t daft4less/noticeboard:v1.0.0 .
[+] Building 50.7s (13/13) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 335B                                0.0s
=> [internal] load metadata for docker.io/library/python:3.11     1.8s
=> [auth] library/python:pull token for registry-1.docker.io      0.0s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 157B                                    0.0s
=> [1/7] FROM docker.io/library/python:3.11@sha256:452f742b83799e5d55163fe5eb5d9b38dd2bcd4f50f69652c3f083fc9bf5 39.2s
=> => resolve docker.io/library/python:3.11@sha256:452f742b83799e5d55163fe5eb5d9b38dd2bcd4f50f69652c3f083fc9bf5 0.0s
=> sha256:bc5f9aa62a1517c80ef76d5a255a16989ecd9bd4f09c0b3ceca6cbb12e9cb332e 249B / 249B 0.2s
=> sha256:3dfcd764bf9335a275461610f56f7748078863cf3843fc3b4b3414218a78a3a 23.98MB / 23.98MB 4.0s
=> sha256:b3085a3e73e9b09a110d7a7ca06e3cd3efa24eb81537alc43c1c748b27629d1d 6.09MB / 6.09MB 3.9s
=> sha256:8d6dd4b254dab2063cd226fca0849d5527402d24d3bea80d644a1e4ac3a47e5 236.18MB / 236.18MB 35.8s
=> sha256:b53089dca50590292ecc77bf803152a5799650e734717e4b706cb812a02073ba 67.78MB / 67.78MB 30.2s
=> sha256:8a7504cd2818ce40ac76c17886a03dffa25ef0aa06ff6125bf0f0c7302cdc6471 25.63MB / 25.63MB 11.2s
=> sha256:f32f49ce655a9cf7c1fd4ca1417ddb39a54cedf4b7ff35de20f8009c18dd7a96 49.31MB / 49.31MB 10.2s
=> => extracting sha256:f32f49ce655a9cf7c1fd4ca1417ddb39a54cedf4b7ff35de20f8009c18dd7a96 0.7s
=> => extracting sha256:8a7504cd2818ce40ac76c17886a03dffa25ef0aa06ff6125bf0f0c7302cdc6471 0.4s
=> => extracting sha256:b53089dca50590292ecc77bf803152a5799650e734717e4b706cb812a02073ba 1.0s
=> => extracting sha256:8d6dd4b254dab2063cd226fca0849d5527402d24d3bea80d644a1e4ac3a47e5 2.8s
=> => extracting sha256:b3085a3e73e9b09a110d7a7ca06e3cd3efa24eb81537alc43c1c748b27629d1d 0.1s
=> => extracting sha256:3dfcd764bf9335a275461610f56f7748078863cf3843fc3b4b3414218a78a3a 0.3s
=> => extracting sha256:bc5f9aa62a1517c80ef76d5a255a16989ecd9bd4f09c0b3ceca6cbb12e9cb332e 0.0s
=> [internal] load build context                                  0.1s
=> => transferring context: 28.80kB                                0.1s
=> [2/7] RUN useradd -m usuario                                  0.6s
=> [3/7] WORKDIR /app                                           0.0s
=> [4/7] COPY requirements.txt .                                0.0s
=> [5/7] RUN pip install -r requirements.txt                     6.9s
=> [6/7] COPY . .                                               0.0s
=> [7/7] RUN chown -R usuario:usuario /app                      0.3s
=> => exporting image                                             1.6s
=> => exporting layers                                             1.2s
=> => exporting manifest sha256:44640ccc605eeaece017c9109fe7db4294e81a921840fe5724e76c0358f712f5f 0.0s
=> => exporting config sha256:724f70cd35439a836219c878583c898db35b36d0e55b3a09e2c75c5cb6f7daf2 0.0s
=> => exporting attestation manifest sha256:8bd90e136bb496c398a7aa4694c1d6309b02d6e97b73ad7be2c69e495f8e3a5a 0.0s
=> => exporting manifest list sha256:55cafe9c9513086d15c91da473b23dd9d402117908cc32d2b5d83deaaf81e083 0.0s
=> => naming to docker.io/daft4less/noticeboard:v1.0.0          0.0s
=> => unpacking to docker.io/daft4less/noticeboard:v1.0.0        0.3s
C:\Users\MSI_KATANA\Downloads\noticeboard\noticeboard>

```

Figura 12: Ejecución del build: Empaquetamiento de la aplicación y dependencias en la imagen Docker.

```
C:\Users\MSI KATANA\Downloads\noticeboard\noticeboard>docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
daf4less/joseweb:latest	b8be1be68caf	92.9MB	26.1MB	U
daf4less/joseweb:v1	b8be1be68caf	92.9MB	26.1MB	U
daf4less/nodeejm:v1	5af6cb26aa7c	193MB	48.4MB	U
daft4less/joseweb:latest	b8be1be68caf	92.9MB	26.1MB	U
daft4less/nodeejm:v1	5af6cb26aa7c	193MB	48.4MB	U
daft4less/noticeboard:v1.0.0	55cafe9c9513	1.67GB	428MB	
danielaag5/dani-web:v1	bfe9330a6afe	92.9MB	26.1MB	U
httpd:latest	bac8021a9b7a	177MB	47.6MB	
joseweb:latest	b8be1be68caf	92.9MB	26.1MB	U
nginx:latest	1881968aff6f	240MB	65.8MB	U
nginx:trixie-perl	3dbffff93f73b	310MB	79.4MB	
nodeejm:latest	5af6cb26aa7c	193MB	48.4MB	U

Figura 13: Verificación de la creación local de la imagen con la etiqueta semántica correcta.

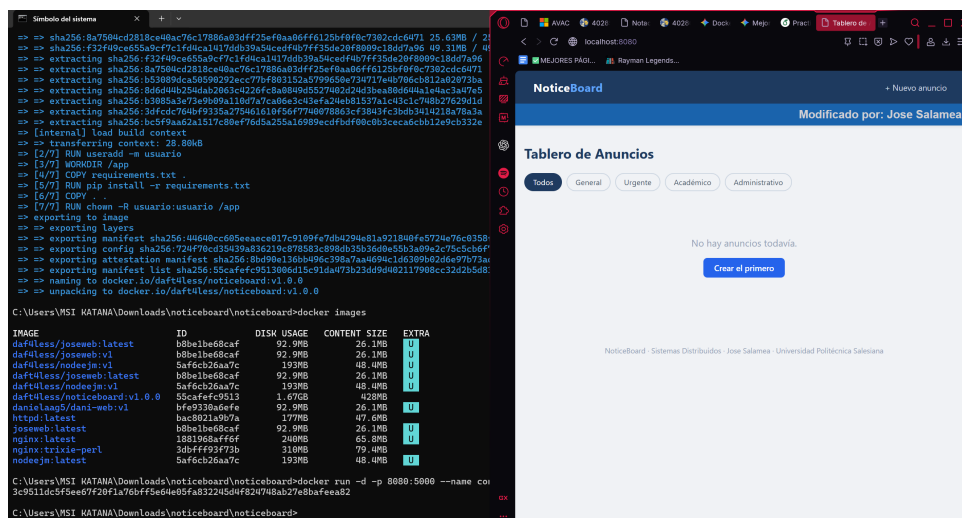


Figura 14: El contenedor respondiendo exitosamente en el puerto mapeado (localhost:8080).

```
C:\Users\MSI KATANA\Downloads\noticeboard\noticeboard>docker logs contenedor_noticeboard
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://172.17.0.1:5000
* Running on http://172.17.0.2:5000
Press CTRL+C to quit
172.17.0.1 - - [22/May/2026 12:30:44] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [22/May/2026 12:30:44] "GET /static/style.css HTTP/1.1" 200 -
172.17.0.1 - - [22/May/2026 12:30:44] "GET /favicon.ico HTTP/1.1" 404 -

What's next:
View and search logs for all containers in one place
with Docker Desktop's Logs view. docker-desktop://dashboard/logs

C:\Users\MSI KATANA\Downloads\noticeboard\noticeboard>
```

Figura 15: Historial de logs generados por Flask demostrando un arranque sin errores dentro del contenedor.

```
C:\Users\MSI KATANA\Downloads\noticeboard\noticeboard>docker exec contenedor_noticeboard whoami
usuario
C:\Users\MSI KATANA\Downloads\noticeboard\noticeboard>
```

Figura 16: Prueba de seguridad validando que el proceso corre bajo el usuario no-root.

3. Publicación en Docker Hub

```
C:\Users\MSI_KATANA\Downloads\noticeboard\noticeboard>docker login
Authenticating with existing credentials... [Username: daft4less]

Info → To login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded

C:\Users\MSI_KATANA\Downloads\noticeboard\noticeboard>docker push daft4less/noticeboard:v1.0.0
The push refers to repository [docker.io/daft4less/noticeboard]
b3085a3e73e9: Pushed
0e9bfb92ad17: Pushed
3dfcdc764bf9: Pushed
8a7504cd2818: Pushed
b53089dca505: Pushed
96f6e856140c: Pushed
ae97cd772e9d: Pushed
a381bd0719b5: Pushed
fdaf3f9abe5f: Pushed
f32f49ce655a: Pushed
8a210ce19b58: Pushed
4a6cfcf1625a: Pushed
8d6d44b254da: Pushed
bc5f9aa62a15: Pushed
v1.0.0: digest: sha256:55cafe9c9513006d15c91da473b23dd9d402117908cc32d2b5d83deanf81e083 size: 856
C:\Users\MSI_KATANA\Downloads\noticeboard\noticeboard>
```

Figura 17: Proceso de inicio de sesión y subida exitosa de las capas al repositorio de Docker Hub.

```
C:\Users\MSI_KATANA\Downloads\noticeboard\noticeboard>docker rmi daft4less/noticeboard:v1.0.0
Untagged: daft4less/noticeboard:v1.0.0
Deleted: sha256:55cafe9c9513006d15c91da473b23dd9d402117908cc32d2b5d83deanf81e083

C:\Users\MSI_KATANA\Downloads\noticeboard\noticeboard>docker run -d -p 8080:5000 --name noticeboard_remoto daft4less/noticeboard:v1.0.0
Unable to find image 'daft4less/noticeboard:v1.0.0' locally
v1.0.0: Pulling from daft4less/noticeboard
fdaf3f9abe5f: Pull complete
96f6e856140c: Pull complete
4a6cfcf1625a: Pull complete
8a210ce19b58: Pull complete
ae97cd772e9d: Pull complete
a381bd0719b5: Pull complete
0e9bfb92ad17: Download complete
Digest: sha256:55cafe9c9513006d15c91da473b23dd9d402117908cc32d2b5d83deanf81e083
Status: Downloaded newer image for daft4less/noticeboard:v1.0.0
45df28bc3471b6f5361bba9519666ff8a7db808541bbd5b22fd775547efd85
C:\Users\MSI_KATANA\Downloads\noticeboard\noticeboard>
```

Figura 18: Descarga y ejecución remota de la imagen tras haber eliminado la versión local.